

less connection. Encryption is done with the network security keys. Access to a key virtually allows access to the particular wireless internet network connection.

There are majorly two types of encryptions used.

WEP (Wired Equivalent Privacy)

This is the most basic form of encryption and hence is the easiest to crack. This is not recommended as it is vulnerable and can be cracked relatively easily. There are uncountable attacks, uncountable vulnerabilities, uncountable research papers listing the issues, and uncountable tools to get the passwords. It isn't hard to learn how to hack these. Despite its shortcomings, many people still stick to this encryption.

WPA (Wi-Fi Protected Access)

This form of encryption is much more secure than the alternative. The only efficient way of cracking a password of such a network is to use a wordlist with the common passwords stored in them. Theoretically, it's a way to get in, and practically this takes forever. Brute force and dictionary attacks are the basic methods to get into the network. A variation of the WPA, WPA-3 is the most secure encryption till now. It is virtually uncrackable with a strong password. The only possibility of cracking it is guessing the paraphrase in case of a weak password, for example, if WPA PIN is still enabled in the router.

Tools Used

We've already dedicated an entire chapter for the most commonly used tools needed for hacking devices and servers. In this chapter, we're more focused on the tools used for cracking your Wi-Fi router's password. The most convenient OS to hack with is Kali Linux. It's a Linux based OS and was specially built for penetration testing purposes. You can download the official version from their website: <https://www.kali.org/downloads>.

Once done with the installation, we need a couple of other basic requirements like a wireless adapter (you're fine if you can connect to the internet using a Wi-Fi), a Wi-Fi network to crack (suggestively your network to avoid legal issues) and few other software.

Aircrack-ng, Wireshark and Reaver are a must to have for hacking a network. Reaver usually comes with Kali Linux. Other software is also discussed in the section below. Download Wireshark as follows: